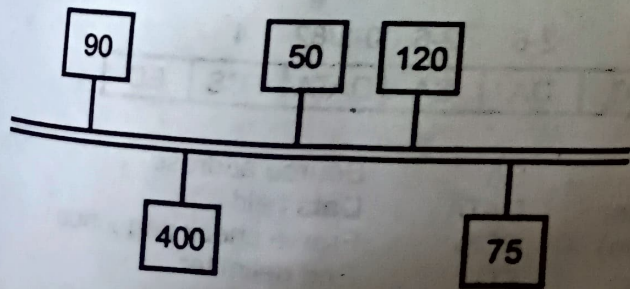


4.6 IEEE 802.4 (TOKEN BUS)

- IEEE 802.4 describes a token bus LAN standard.
- The IEEE 802.4 standard for media access control is known as Token bus.
- Token bus is a linear or the tree shaped cable through which different stations are interconnected.
- In token passing method stations, connected on a bus are arranged in a logical ring. When the logical ring is initiated, the highest numbered station may send the first frame. After this it passes permission to its immediate neighbor by sending a special control frame called a token. The token propagates around the logical ring, with only the token holder being permitted to transmit frames. Since only one station at a time holds the token, collisions do not occur. There is no relation between the physical location of the station on the bus and its logical sequence number. Fig. 4.13 illustrates the operation of token bus.

Physical topology



Logical sequence of token passing

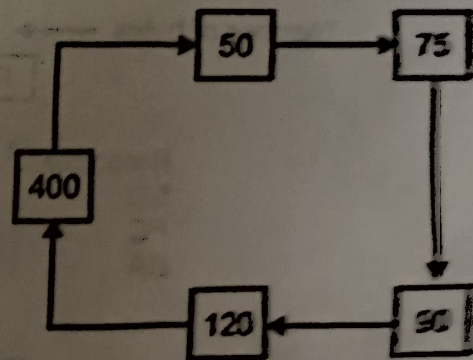


Fig. 4.13 Token passing sequence in a bus

4.6.1 Features of IEEE 802.4

1. It uses highly reliable cable television equipment.
2. It is more deterministic than 802.3
3. It can handle short minimum frame.
4. It also supports priorities.
5. Token bus has excellent throughput and efficiency at high load.
6. Token bus supports good bandwidth to high priority traffic.

4.6.2 IEEE 802.4 Cable Standards

- The token bus standard specifies three physical layer options in terms of transmission medium, signaling technique, data rate and maximum electrical cable segment length. Table 4.5 summarizes these options defined for the IEEE 802.4 medium.

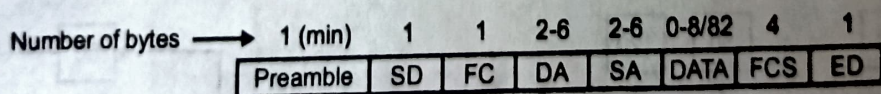
Table 4.5 IEEE 802.4 medium options

Sr. No.	Medium Options	Transmission Medium	Signaling Technique	Data rate (Mbps)	Maximum segment length (m)
1.	Broad band	Coaxial cable (75 ohm)	Broad band (AM/PSK)	1, 5, 10	Not specified
2.	Carrier band	Coaxial cable (75 ohm)	Broad band (FSK)	1, 5, 10	7600
3.	Optical fiber	Optical fiber	ASK Manchester	5, 10, 20	Not specified

1. **Broad band.** Broadband system supports data channel at 1, 5, 10 Mbps with bandwidths of 1, 5, 6 and 12 MHz, respectively. The standard recommends the use of a single cable split system with a head end frequency translator.
2. **Carrier band.** It is also known as **single-channel** broadband. Because carrier-band is dedicated to a single data channel, it is not necessary to take care that the modem output is confined to a narrow bandwidth.
3. **Optical fiber.** It is the most recent option at data rates of 5, 10 and 20 Mbps. Either a passive or active star topology can be used.

4.6.3 Frame Format of IEEE 802.4

The frame format as specified by IEEE 802.4 is as shown in Fig. 4.14



Preamble	: Bit synchronization	SA	: Source address
SD	: Frame Start Delimiter	DATA	: Data field
FC	: Frame Control (Type)	FCS	: Frame check sequence
DA	: Destination address	ED	: End delimiter

Fig. 4.14 Format of IEEE 802.4 frame

The frame consists of following fields:

- (i) Preamble. Preamble is at least one octet long and used for bit synchronization.
- (ii) Start Delimiter (SD). It is a unique one-byte pattern, which marks the beginning of a frame.
- (iii) Frame Control (FC). The field indicates the type of frame. It is one octet long and indicates if the frame is a data frame or control frame. Token is one of the control frames.
- (iv) Source Address (SA). The source address field is also 2 or 6 octets long.
- (v) Destination Address (DA). It contains the destination address and it is 2 to 6 bytes long.
- (vi) Frame Check Sequence (FCS). This field contains a CRC code. It is 4 bytes long and used to check on DA, SA, FC and Data fields.
- (vii) End Delimiter (ED). This is one byte unique bit pattern, which marks the end of the frame. The total length of the frame from FC to FCS field including the Data field can be at the most 8191 octets.

4.6.4 Performance of IEEE 802.4

- or Token ring, the slightly higher delay compared to CSMA/CD bus occurs. For higher transmission loads the token ring performs well.

4.6.5 IEEE 802.5 (Token Ring)

- In a token ring a special bit pattern, called the token, circulates around the ring whenever all stations are idle. When a station transmits, it breaks the ring and inserts its own frame with destination and source addresses. When the frame eventually returns to the originating station after completing the round, the station removes the frame and closes the ring. Because there is only one token, only one station can transmit at a given instant, thus solving the channel access problem. Fig. 4.15 shows token ring arrangement.
- Each station is connected to the ring through a Ring Interface Unit (RIU). The sequence of token is determined by the physical locations of the stations on the ring.

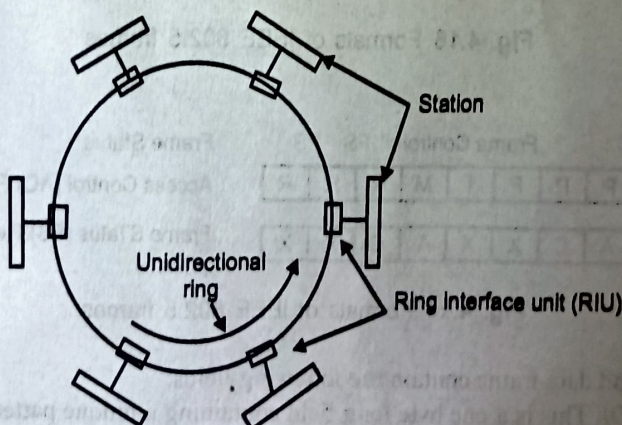


Fig. 4.15 A ring network

4.6.6 Features of IEEE 802.5

1. It uses point-to-point connection.
2. Throughput and efficiency at high load are excellent.
3. Token ring support short frames.

4.6.7 IEEE 802.5 Cable Standards

The token ring standard IEEE 802.5 specifies two types of transmission medium discussed below.

1. **Shielded twisted pair (STP)**. It uses differential Manchester signaling technique. Data rate 16 Mbps. Maximum number of repeaters allowed is 250.
2. **Unshielded twisted pair (UTP)**. It uses differential Manchester signaling technique. Data rate 10 Mbps; maximum number of repeaters allowed is 250.

4.6.8 Frame Format of IEEE 802.5

The IEEE 802.5 has standardized the formats for the token frame and data frame. They are as shown in Fig. 4.16.

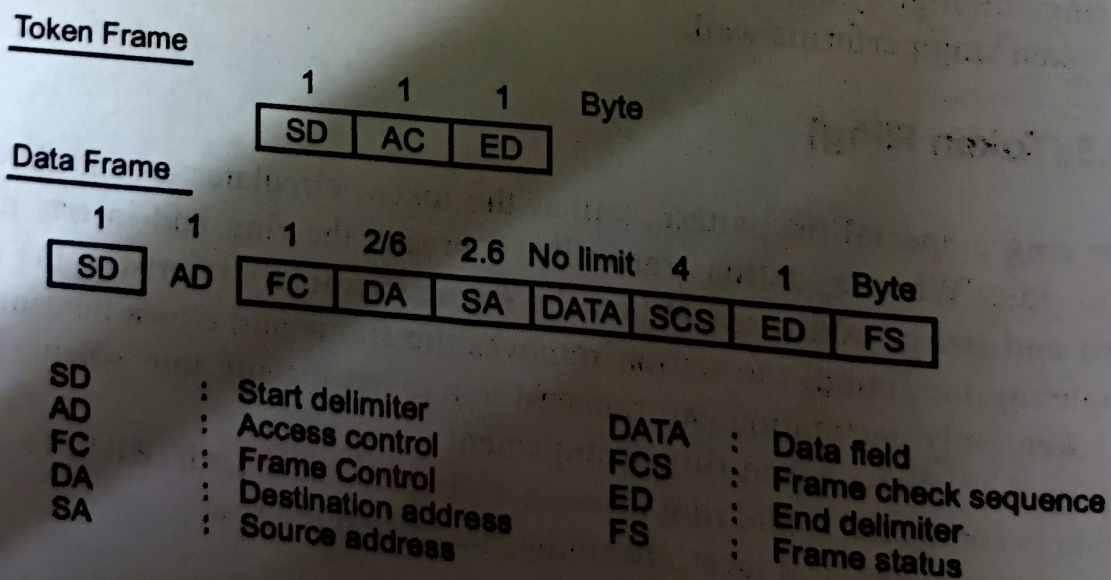
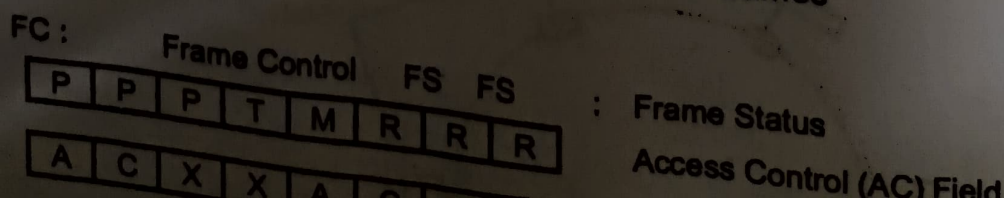


Fig. 4.16 Formats of IEEE 802.5 frames



- **Routers:** Routers connect network segments, and retain its own independent best path from one network to another network, and forward packets to the destination.
- **Gateways:** Gateways operate at higher layers of the protocol stack; some gateways operate on all layers. Gateways, also called protocol converters, connect different networking environments, such as Systems Network Architecture (SNA) and Transmission Control Protocol/Internet Protocol (TCP/IP) networks, by converting each header from the form of the sending environment to that of the receiving environment.
- **Brouters:** A brouter can work in either the second and third layers of the OSI model-the data link layer or the network layer. A brouter is a combination of a bridge and router combined. If it can't route a packet, it acts as a bridge.

4.9.1 Repeaters

Repeaters working at the Physical Layer, are the simplest type of internetworking device. They are used to amplify electrical signals carried by the network. The function of a repeater is to receive incoming signals (a packet of data), regenerate the signals to their original strength, and retransmit them. Repeaters are used to lengthen individual network segments to form a larger extended network. That is, repeaters allow a network to be constructed that exceeds the size limit of a single physical segment by allowing additional lengths of cable to be connected (see figure 4.22).

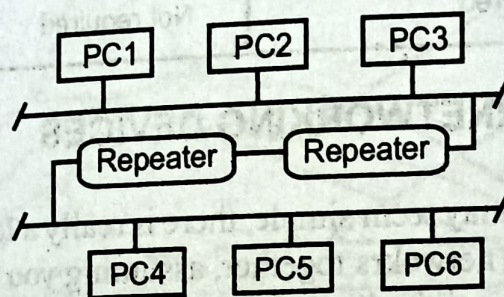


Fig. 4.22 Repeaters used to lengthen individual network segments

4.9.1.1 How Repeaters Work?

As nodes are added to a LAN, the cable can be extended with electrically passive connectors that simply attach one segment to another. Eventually, the permissible length of the cable segment will be exceeded, or it will become necessary to connect different cable types. A repeater is used to address these needs.

Because a repeater operates at the Physical Layer of the OSI model, the job of a repeater is to repeat bits. If a "1" bit is received on the input port of a repeater, a "1" bit is regenerated, with a stronger signal at the output of the repeater. If a "0" bit is received on the input port of a repeater, a "0" bit is regenerated at the output of the repeater. A repeater is considered a **non-discriminating** device, because all incoming signals are passed on to each connected segment. Because a repeater reproduces exactly what it receives bit-by-bit, it can **reproduce errors**. However, repeaters are **very fast** (10 megabits per second [Mbps] for Ethernet) and cause very little delay.

As internetworking devices for Ethernet LANs, repeaters are feasible only for relatively small LANs (less than 100 nodes), confined to a small geographical area such as one or two floors of an office building. A repeater should not be used to connect heavily used LANs because it cannot isolate traffic between LAN segments. Because each bit is copied to the attached segments, all data passes through a repeater in both directions. Therefore, if we connect multiple LAN segments using a repeater, we may experience performance problems, because total network traffic will increase.

4.9.2 Hubs

Hubs are argued as not being real internetworking devices. The reason we are including them in the internetworking portion is that hubs are often incorporated into other internetworking devices in order to minimize the amount of equipment installed at customer sites.

A hub is a box with a number of connectors to which multiple nodes (PCs) are attached. It serves as a common termination point that can relay signals along the appropriate paths. All hubs provide connectivity and some even provide management capabilities. A hub usually connects nodes that have a common architecture.

Like repeaters, hubs operate at the Physical Layer, and regenerate signals bit-by-bit as they travel down the physical medium. Both repeaters and hubs provide signals greater reach; however, a hub also shares the

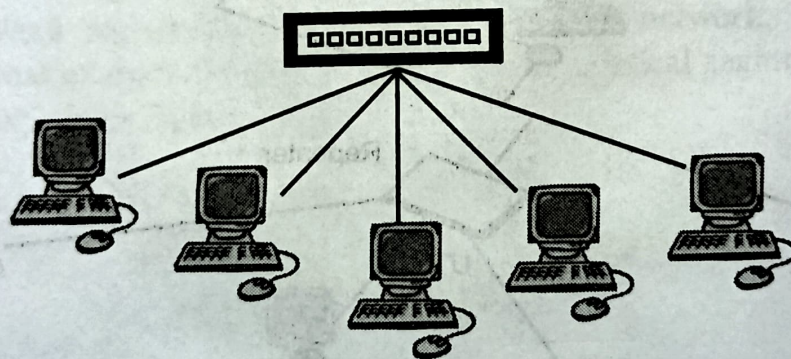


Fig. 4.25 Hub in a star topology

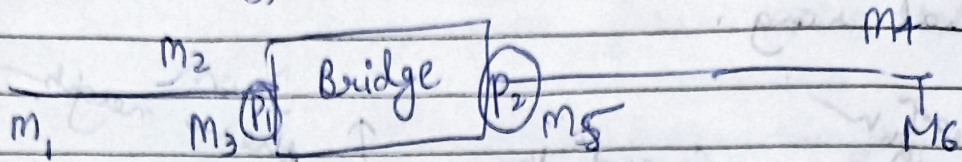
same signals among a group of nodes. This clustering helps control the morass of cable and cable types found in many networks.

The Hub in figure 4.25 illustrates a simple hub configuration. It is important to realize that this structured wiring distinguishes between a network's physical and logical topologies. All hubs use a physical star topology; however, the logical topology of the network segment can be a ring, bus, tree or other structure depending on the way a signal is passed from one node to the others.

connects different LANs

BRIDGE

- (1) Connects diff LAN'S



MAC ADDRESS

Source	Address
--------	---------

- (2) Can check MAC address to decide where to send.
- (3) It can forward packets

Types

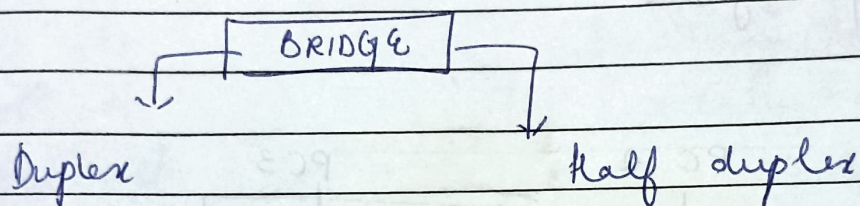
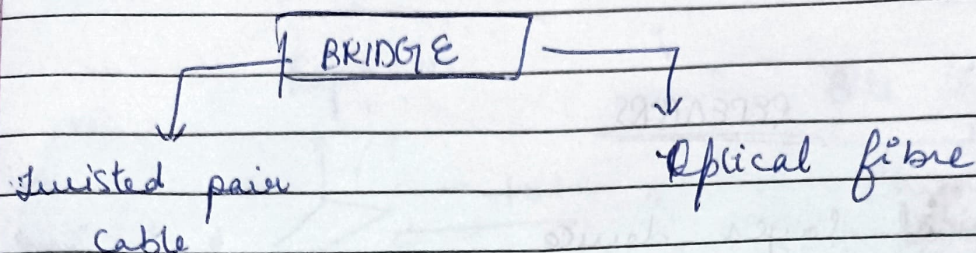
Static

Dynamic

Mac	Port address
m_1	P_1
m_2	P_1
m_3	P_1
m_4	P_2
m_5	P_2
m_6	P_2

If a frame that is meant for m_2 arrives at port 2, the bridge will check its table to send the frame through port mentioned in the table.

Bridge is used to connect mediums of different types



ADVANTAGES

- ① uses optimal route (spanning).

DISADVANTAGE

FILTERING Bridge has filtering capacity

It can check mac address to decide if a packet can be dropped.

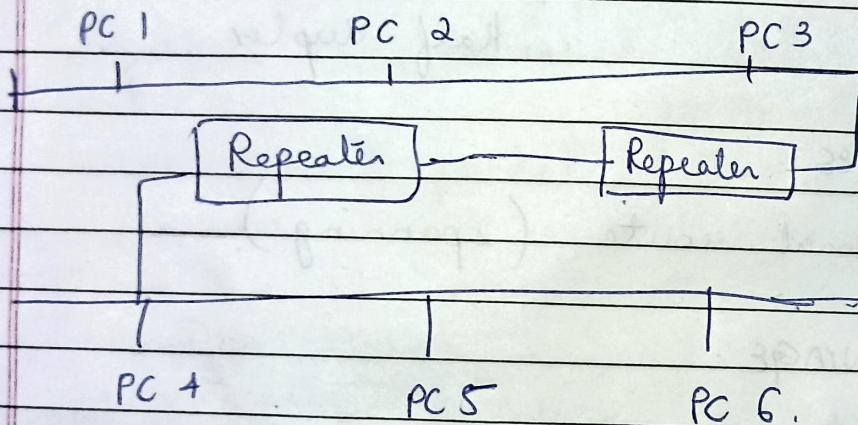
TYPES

- ① TRANSPARENT BRIDGE: The existing stations have no idea that a bridge exists. Addressing table resides in memory.
- ② ROUTING BRIDGE: A station that sends the data, defines the bridge that the

frame will visit. In routing bridge, frame contains source, destination & bridge address.

REPEATERS

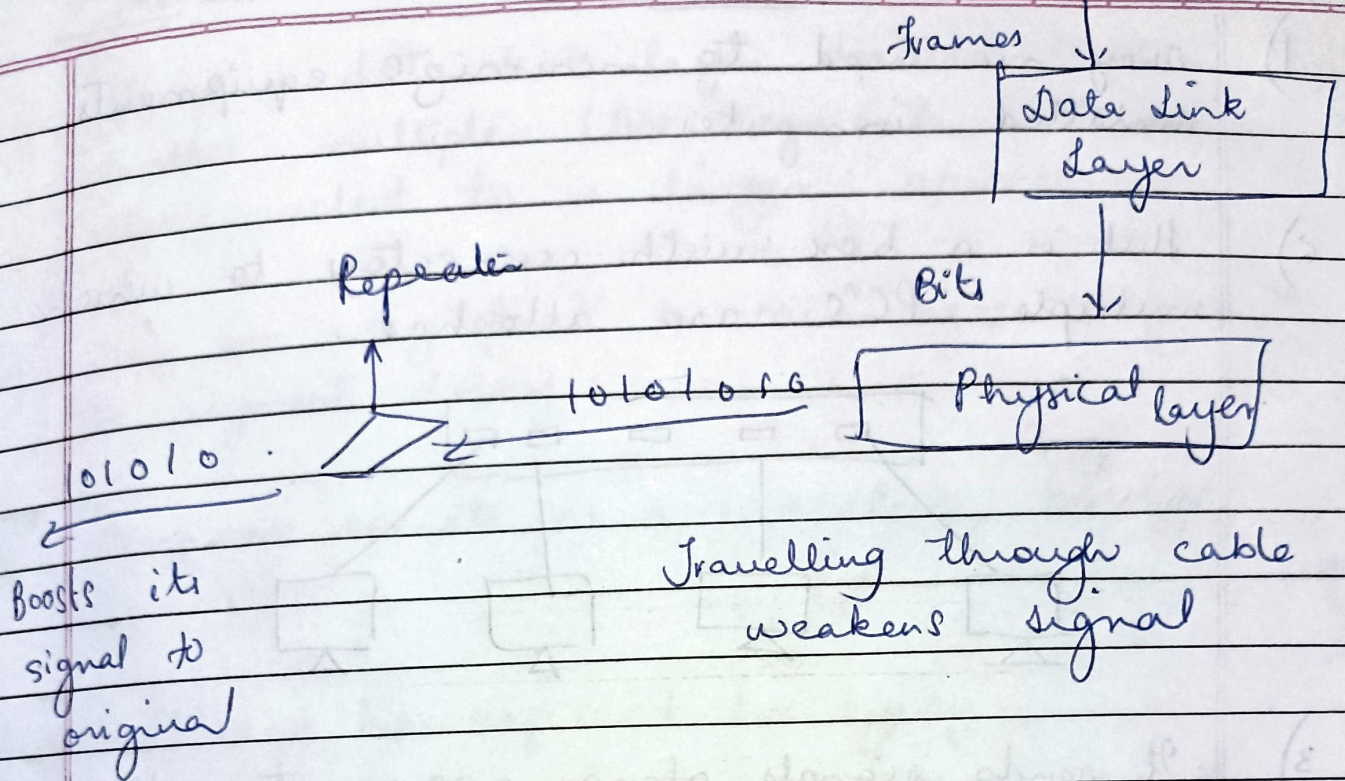
- 1) Physical layer device
- 2) amplify signals



- 3) Regenerates signal to its original strength before retransmission
- 4) Lengthens network segments to form a large network
- 5) Non discriminating device, reproduces errors also

ADVANTAGES

- 1) Fast, causes no delay
- 2) Can connect different media types
~~IEEE 802.3~~ Ethernet & Token ring

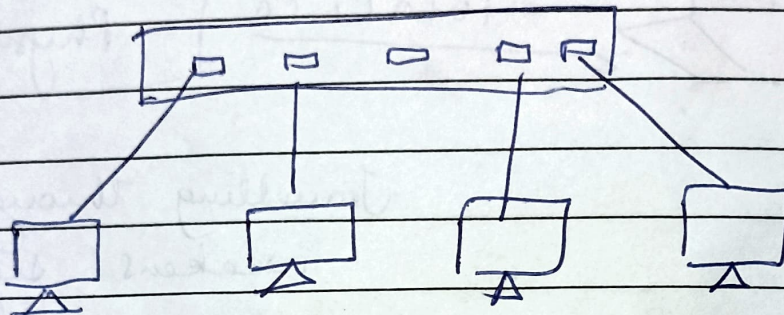


DISADVANTAGES:

- 1) Used for small LANS (100 nodes)
 - 2) Covers short geographical area one floor
 - 3) Cannot be used for heavy traffic n/w
 - 4) Data is copied to all segments
- Right ← ————— → left
- which results in increased traffic

HUBS

- 1) They are used to minimize equipments installed in sites
- 2) Hub is a box with connectors to which multiple PC's are attached.



- 3) It sends signals along appropriate paths.
- 4) Hubs provide connectivity some even ^{have} ~~show~~ management capabilities.
- 5) Connects nodes having common architecture.
- 6) Work at physical level, Reduces the mess of cables.
- 7) Hubs use star topology, but logical topology can be ring or bus --

SWITCHES

- 1) It is a data link layer device that enables multiple LAN segments to be interconnected to a larger n/w.
- 2) They are very fast & minimize segment to segment delay.
- 3) They can operate as a repeater, bridge or router.
- 4) Switching has replaced bridging.
- 5) Divides n/w into self contained units can manage traffic.

ADVANTAGE

- 1) High throughput performance
- 2) Lower per port cost
- 3) Greater flexibility
- 4) Acts as a firewall & prevents n/w error